As is the case for Newton iteration, one may use the constraint residuals as an additional test. It is certainly unsafe to check only the latest difference of iterates (of a single component), and one should instead check that these are small for an entire cycle. The iteration described above is an example of a Gauss-Seidel-Newton iteration and it and many other iterative nonlinear solvers are discussed in [292].

4.3.4 Symplectic Constrained Dynamics Integrators

Is (4.14)–(4.16) symplectic (in the sense outlined above for holonomic systems)?

It is difficult to make sense of this question, since (4.14)–(4.16) does not even constitute a map of the co-tangent bundle. On the other hand it is not too difficult to correct this defect by incorporating an additional projection onto the cotangent space:

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}_{n+1}, \tag{4.20}$$

$$\bar{\boldsymbol{p}}_{n+1} = \boldsymbol{p}_n - h\boldsymbol{F}_n - h\boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_n, \tag{4.21}$$

$$\boldsymbol{0} = \boldsymbol{g}(\boldsymbol{q}_{n+1}), \tag{4.22}$$

$$\boldsymbol{p}_{n+1} = \bar{\boldsymbol{p}}_{n+1} - \boldsymbol{g}'(\boldsymbol{q}_{n+1})^T\boldsymbol{\mu}_{n+1}, \tag{4.23}$$

$$\boldsymbol{0} = \boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1}. \tag{4.24}$$

The demonstration that the method is symplectic can be seen as a consequence of the following lemma.

Lemma 4.1 *Let $\gamma : \mathbb{R}^{N_c} \to \mathbb{R}$ be C^2. Let $\boldsymbol{q}, \boldsymbol{p} \in \mathbb{R}^{N_c}$ be constrained to the manifold defined by*

$$\gamma(\boldsymbol{q}) = 0, \qquad \boldsymbol{p} \cdot \nabla\gamma(\boldsymbol{q}) = 0,$$

and suppose $\boldsymbol{P}, \mu$ satisfy

$$\boldsymbol{P} = \boldsymbol{p} - \mu\nabla\gamma(\boldsymbol{q}).$$

Then

$$\sum_{j=1}^{N_c} \mathrm{d}q_j \wedge \mathrm{d}P_j = \sum_{j=1}^{N_c} \mathrm{d}q_j \wedge \mathrm{d}p_j.$$